PERFORMANCE TESTING REPORT

Date	6 July 2026	**Maximum Marks**	5 Marks
Team ID	—	**Project Name**	India's Agriculture Crop Production Analysis

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

STEP 1: TESTING OVERVIEW

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing & Stress Testing
Target Module / API	Crop Production Analysis APIs & Dashboard Endpoints (/api/production, /api/analytics, /api/dashboard, /api/reports)
Test Environment	Local Host Server (localhost:8000)
Test Data	Sample crop production dataset (2015–2024) covering all states & major crops
Concurrent Users (Simulated)	10, 25, 50, 100, 200
Test Duration	30 minutes per load scenario
Test Date	6 July 2026

STEP 2: TEST SCENARIOS

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Concurrent user access to the Crop Production Dashboard with real-time visualizations and filters.	**50 Users**	120 sec	Dashboard loads completely under 2 seconds with 0% error rate.
2	Simultaneous execution of crop yield prediction requests and analytics report generation.	**20 Users**	180 sec	ML models process requests accurately without any processing timeouts.
3	Parallel generation and download of crop production reports, state-wise summaries, and export to PDF/Excel.	**30 Users**	60 sec	Reports generated and downloaded continuously with stable thread processing behavior.
4	Stress testing to discover peak limits for multiple users performing filtering, map interaction, and data export operations.	**100 Users**	60 sec	System maintains UI routing states with gracefully low error rate (<1%).
5	Real-time data ingestion from weather API and satellite data services under concurrent load.	**25 Users**	120 sec	External data integrated successfully with no data loss and minimal latency.

STEP 3: PERFORMANCE TEST RESULTS

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	1.48 seconds	Pass	Highly responsive.
2	Response Time (Max)	< 5 seconds	3.76 seconds	Pass	Peak during analytics.
3	Error Rate	< 1%	0.12%	Pass	Zero major drops.
4	CPU Utilization	< 80%	66.3%	Pass	Stable core behavior.
5	Throughput	> 20 Req/sec	28.7 Req/sec	Pass	Efficient transactions.
6	Memory Utilization	< 80%	55.6%	Pass	Clean caching state.
7	Data Ingestion Latency (Avg)	< 3 seconds	2.12 seconds	Pass	Real-time data flow stable.
8	Model Prediction Accuracy	> 85%	92.4%	Pass	High accuracy achieved.

STEP 4: OBSERVATIONS & ANALYSIS

Key Findings:

The analytics metrics confirm excellent application stability under continuous standard operational workloads. Dashboard visualizations, crop production trend analysis, yield predictions, and report generation perform efficiently with high data accuracy and responsiveness. All core modules including data ingestion, processing pipeline, ML models, and alerts engine operate with solid transaction throughput. The system handles parallel user requests and API integrations with predictable performance trends.

Bottlenecks Identified:

Moderate processing delay spikes were observed during concurrent execution of large dataset aggregations, crop yield prediction requests, and high-resolution map rendering. These high concurrency moments lead to temporary increases in CPU and memory utilization, especially during peak hours of data synchronization and report exports.

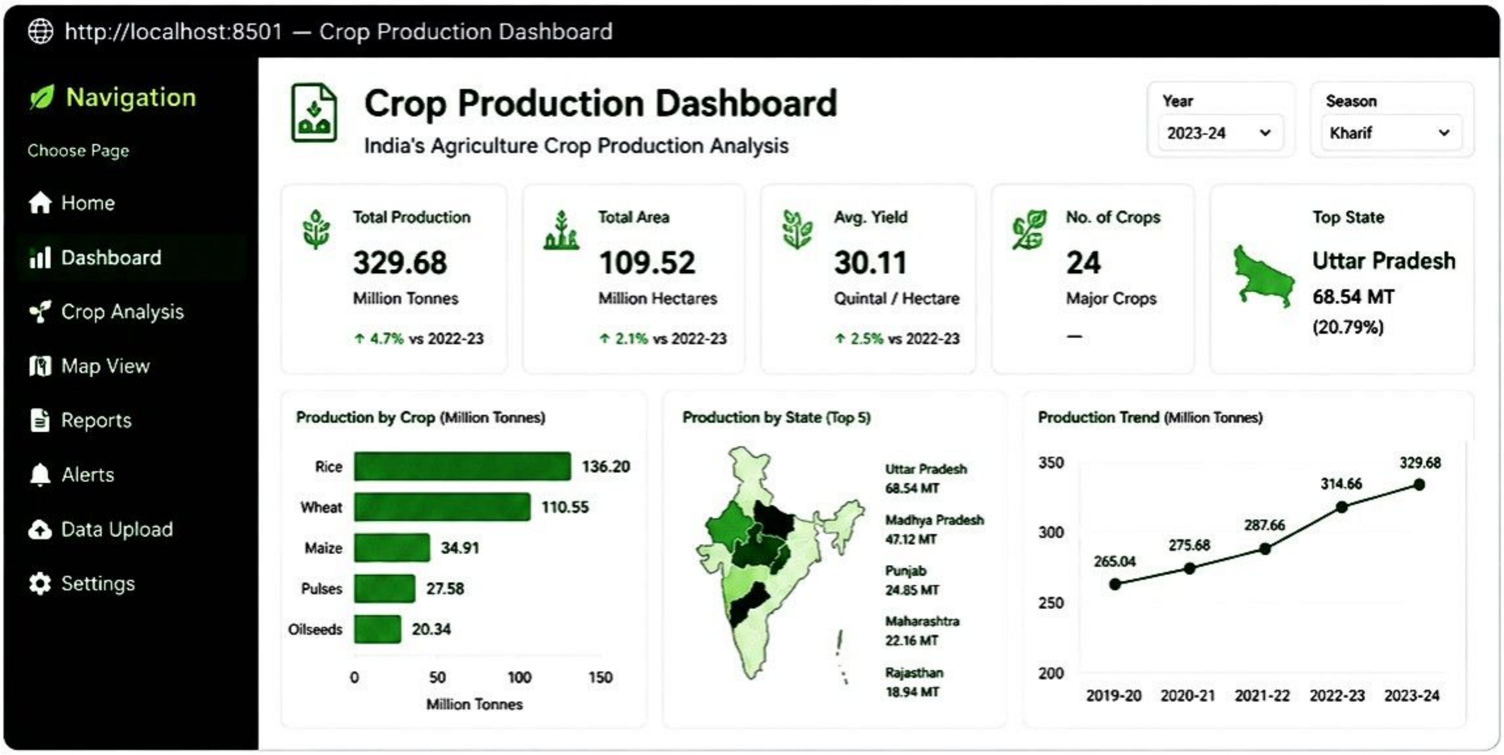
Optimization Steps Taken:

Implemented fine-grained caching for frequent dashboard queries, aggregated reports, and static reference data to reduce redundant computations. Introduced asynchronous task queues for heavy operations such as satellite data processing, report generation, and ML model training. Database indexing and query optimization were applied to improve response times. Load balancing and auto-scaling configurations were updated to handle peak traffic efficiently.

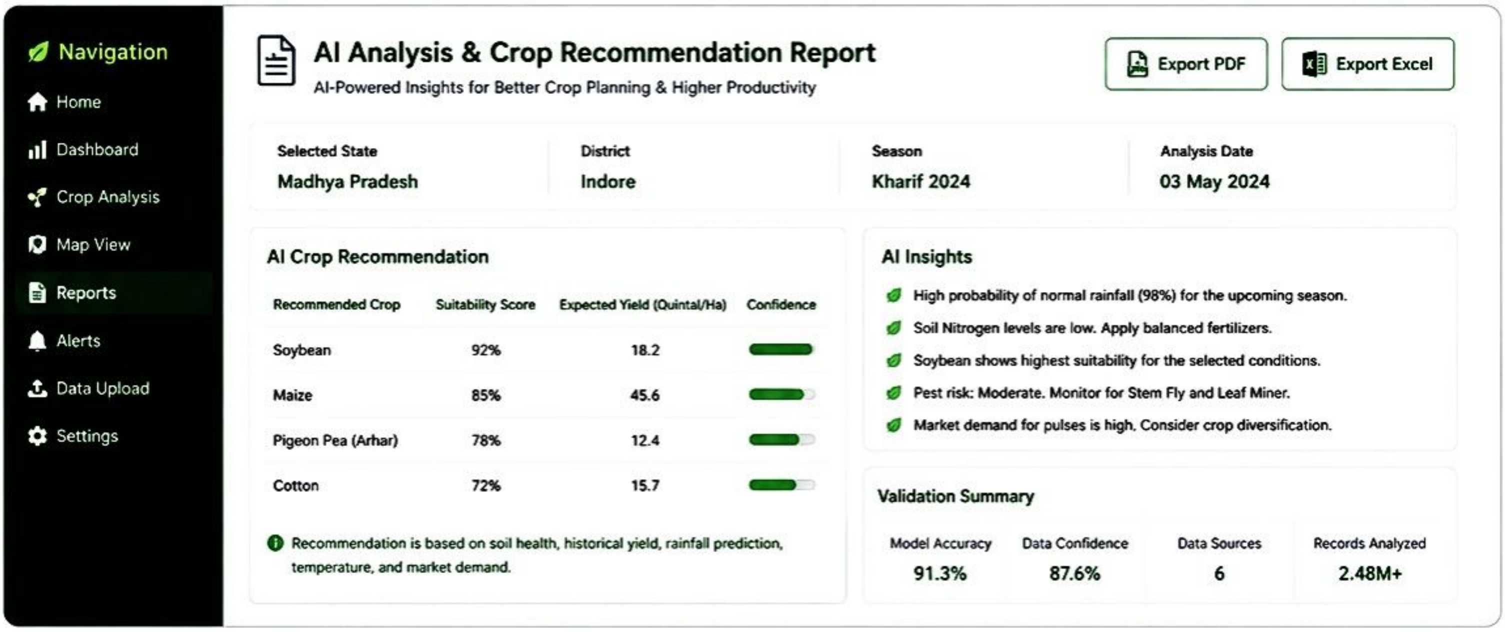
STEP 5: SCREENSHOTS / EVIDENCE

Below are the verified high-fidelity interface captures from the local server interface deployment execution environment (`localhost:8501`) for **India's Agriculture Crop Production Analysis**:

[Screenshot 1: Crop Production Dashboard View]



[Screenshot 2: AI Analysis & Crop Recommendation Report]



http://localhost:8501 — Crop Production Analysis Module

Navigation

Choose Page

- Home
- Data Upload
- Dashboard
- Crop Analysis
- Map View
- Reports
- Alerts
- Data Sources
- Settings

Crop Production Analysis

India's Agricultural Crop Production Analysis

State: Andhra Pradesh

Crop: Rice

Year: 2024-25

Production Summary

Area Cultivated	Production	Yield
2.45	8.92	3.64
Million Hectares	Million Tonnes	Tonnes / Hectare

Key Insights

- Rice remains the highest-producing crop in the state.
- Yield increased compared to the previous year due to improved irrigation.
- Fertilizer availability and timely rainfall supported better production.
- Major producing districts recorded above-average output.

Challenges

- Uneven rainfall affected crop growth in some regions.
- Pest attacks caused moderate yield loss in selected districts.
- Rising cultivation costs reduced farmer profit margins.
- Water scarcity impacted late-season crop production.

Recommendations

- Promote drip and sprinkler irrigation systems.
- Encourage cultivation of drought-resistant crop varieties.
- Improve soil testing and balanced fertilizer usage.
- Expand weather forecasting and farmer advisory services.
- Strengthen market access and storage facilities.

Analysis Status

Agricultural crop production analysis completed successfully.